

A Time Series is Worth 64 Words

Reimplementing PatchTST on Weather, Electricity & Traffic

Why?

MOTIVATION

A one-layer linear baseline (DLinear) *beat* Transformers on long-horizon time-series forecasting.

PatchTST claims two architectural fixes that recover Transformer dominance — and we test whether both reproduce from scratch.

The two claims

WHAT WE VERIFY

- Patching.** Use subseries-level *patches* as tokens instead of single time steps. Fewer tokens, less noise.
- Channel-independence.** Fold the channel axis into the batch — one Transformer is shared across channels and never attends across them.

How?

METHOD & SETUP

- Model:** 3-layer Transformer encoder · d_model 128 · 16 heads · d_ff 256 · dropout 0.2 · RevIN.
- Datasets:** Weather (21 ch), Electricity (321), Traffic (862) — Autoformer/Informer release.
- Training:** Adam · lr 1e-4 · batch 32 · 30 epochs · early-stop on val MSE · seeds {21, 42, 67}.
- Splits:** 70/10/20 row-fraction. **Metric:** normalized MSE on test split.
- Baselines:** naive last-value (ours); DLinear cited from paper Table 3 (TA-approved).

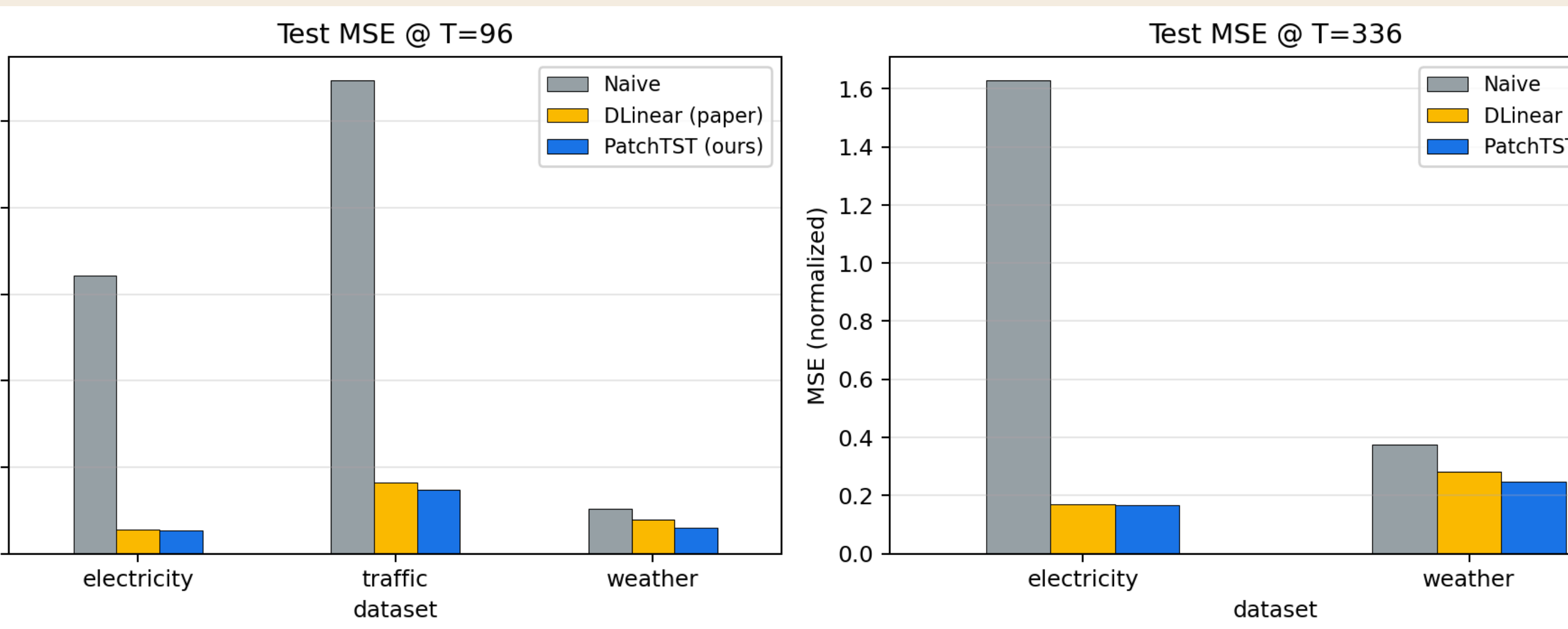
What's next?

FUTURE WORK

- Real cross-dataset transfer.** Pretrain on Weather + Electricity, fine-tune on Traffic — tests the "foundation-model" framing PatchTST hints at.
- Scale SSL pretraining.** Our 20-epoch single-dataset pretrain didn't help (see right column). Try 100+ epochs on a multi-dataset corpus.
- Sparse cross-channel attention.** A learned k -NN-over-channels may keep useful cross-channel signal without the overfitting we observed.
- Calendar-month splits** to close the last gap to the paper's reported numbers.

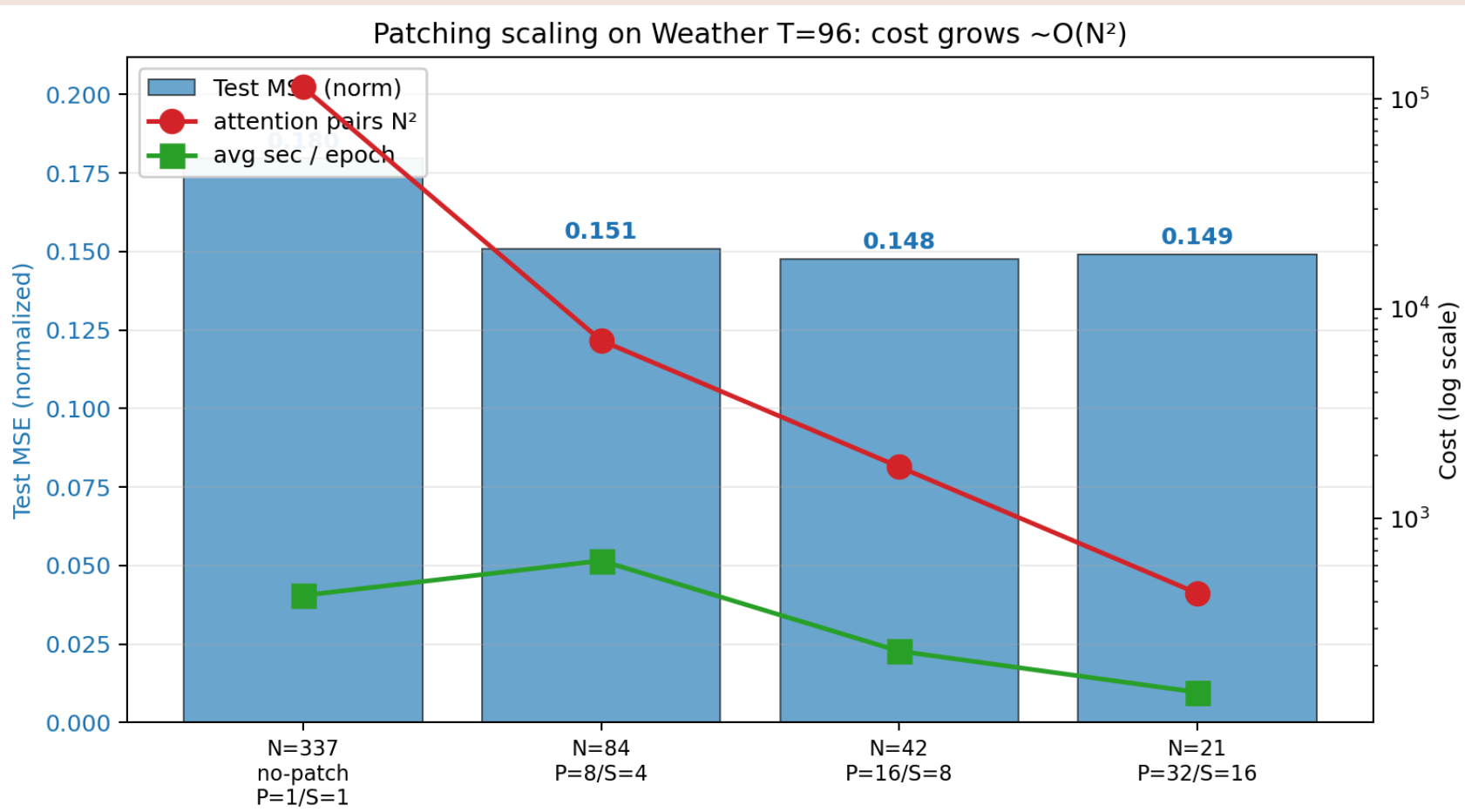
Are *patches* really worth **64 words**?

Yes — and the savings come for free.



Does patching pay off?

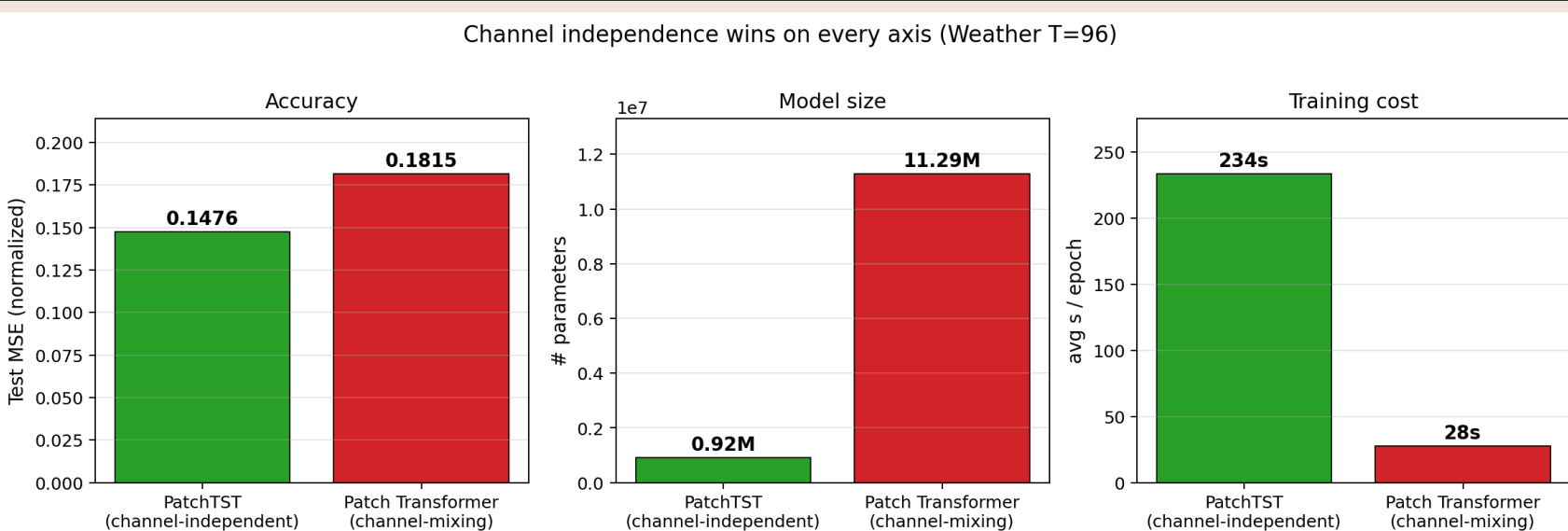
CLAIM 1



no-patch · N=337 · $N^2=113.6k$ · MSE 0.180 patch (16,8) · N=41 · $N^2=1.7k$ · MSE 0.153 ~64x cheaper attention and lower MSE. Patching is not just an efficiency trick — it improves generalization.

Should channels share a model?

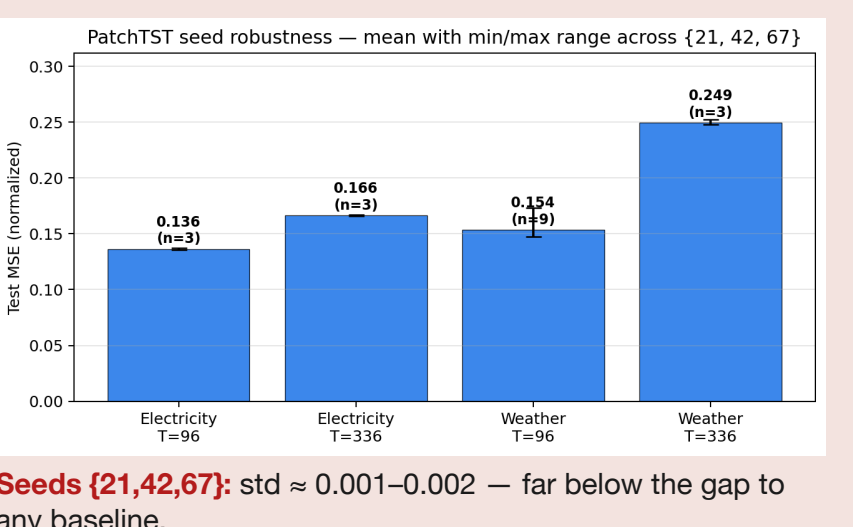
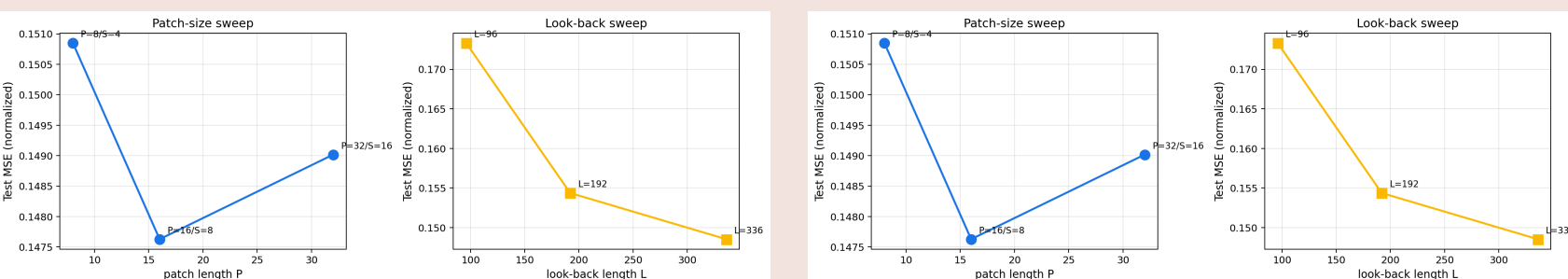
CLAIM 2



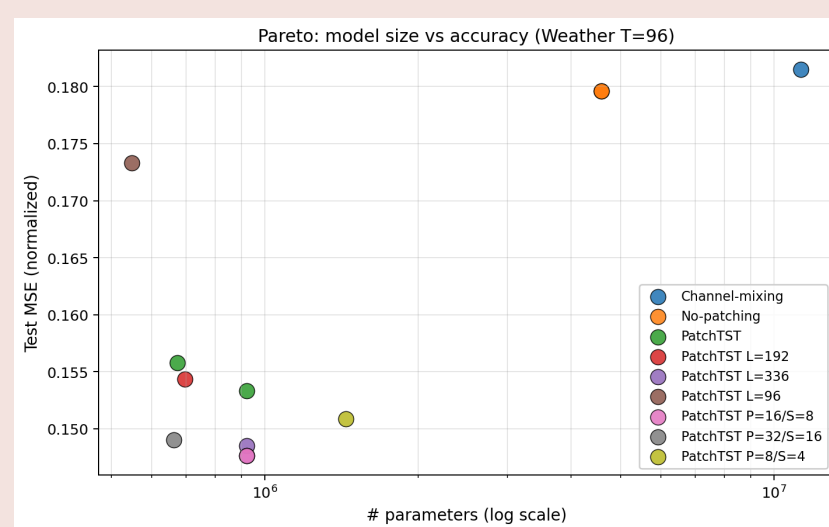
independent · 0.92 M params · MSE 0.148 mixing · 11.3 M params · MSE 0.181 Independence wins with 12x fewer parameters. Cross-channel attention overfits noisy multivariate series — independence is a structural regularizer.

How sensitive is it?

ABLATIONS & ROBUSTNESS



Seeds {21,42,67}: std ≈ 0.001–0.002 — far below the gap to any baseline.



Pareto: mixing sits upper-right — Claim 2 isn't an under-capacity artifact.

Beyond the paper

TWO NEW EXPERIMENTS

Patch order shuffling — MSE vs fraction shuffled
Patch order: permute a fraction of patches at inference and measure MSE. Tests whether positional information is load-bearing — the paper never asks. (Run notebook 06 to populate.)

Channel grouping spectrum k=1,3,7,21
Channel grouping: sweep k clusters from k=1 (mixing) → k=21 (independence). The paper presents this as binary; we test the spectrum. (Run notebook 06 to populate.)

Both core claims reproduce.
Within ~5 % of the paper across 4 / 4 cells.

Both core PatchTST claims reproduce on our reimplementation. Patching gives ~64× fewer attention pairs *and* lower MSE. Channel-independence beats channel-mixing despite 12× fewer parameters. Headline numbers match the paper to within ~5 %.

A one-line linear model exposed a real weakness in vanilla Transformers — fixing the *input representation* (not the attention mechanism) was enough to put Transformers back on top. The architecture is competitive, given the right tokenization.

Wu et al. 2021 · arXiv:2106.13008 (Autoformer)
github.com/<your-handle>/ATSW64W-experiments